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Magnetic energy-based feature extraction for low-quality fingerprint images

Ahmad B. A. Hassanat[1] · V. B. Surya Prasath[3,4,5] · Mouhammd Al-kasassbeh[1] · Ahmad S. Tarawneh[1,2] · Ahmad J. Al-shamailh[1]

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Abstract
In fingerprint recognition systems, feature extraction is an important part because of its impact on the final performance of the overall system, particularly, in the case of low-quality images, which poses significant challenges to traditional fingerprint feature extraction methods. In this work, we make two major contributions: First, a novel feature extraction method for low-quality fingerprints images is proposed, which mimics the magnetic energy when attracting iron fillings, and this method is based on image energies attracting uniformly distributed points to form the final features that can describe a fingerprint. Second, we created a new low-quality fingerprints image database to evaluate the proposed method. We used a mobile phone camera to capture the fingerprints of 136 different persons, with five samples for each to obtain 680 fingerprint images in total. To match the computed features, we used the dynamic time warping and evaluated the performance of our system based on k-nearest neighbor classifier. Further, we represent the features using their probability density functions to evaluate the method using some other classifiers. The highest identification accuracy recorded by several experiments reached 95.11% using our in-house database. The experimental results show that the proposed method can be used as a general feature extraction method for other applications.

Keywords Fingerprint recognition · Low-quality images · Feature extraction · Magnetic energy · Smartphone camera

1 Introduction

Driven by its high accuracy, low cost, and acceptability, fingerprint recognition has become one of the most famous biometric systems recently, since it is used in highly sensitive applications such as banks, airports, border crossings, cellular phones, personal computers, laptops, and other systems that consider information security a key issue [3]. This biometric trait could also be used in criminology-related security in the same way, and most of law agencies in the world are routinely collecting fingerprints of apprehended criminals to track their criminal history [20]. In recent years, tremendous efforts of research have been made with fingerprint recognition [11–13,28]. These systems depend mainly on extracting features from fingerprint images, which are captured using specific scanners; however, these scanners are generally expensive and are unavailable for most users [6]. Further, research on fingerprint systems has achieved significant accomplishments in terms of efficiency, and many of these recognition systems require high resolution images. Therefore, many researchers preferred to capture fingerprint images using low-quality images, such as those captured by a commercial cameras [8,14,29], smartphones /mobile phone cameras [15,16,22,23,26], and webcams [10,18].

Most of the previous studies used standard databases or created new databases for the purposes of evaluating the performance of their systems. In terms of feature extraction, the common approaches targeted any of the following: (1)

✉ V. B. Surya Prasath
prasaths@missouri.edu

1 Department of Information Technology, Mutah University, Karak, Jordan

2 Department of Algorithms and Their Applications, Eötvös Loránd University, Budapest, Hungary

3 Computational Imaging and VisAnalysis (CIVA) Lab, Department of Computer Science, University of Missouri-Columbia, Columbia, MO 65211, USA

4 Present Address: Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center, Cincinnati, OH 45229, USA

5 Department of Biomedical Informatics, College of Medicine, University of Cincinnati, Cincinnati, OH, USA

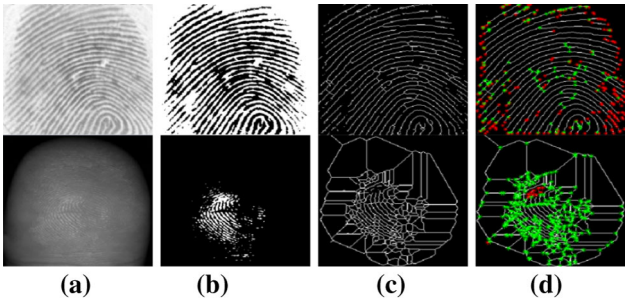

Fig. 1 Low-quality fingerprints are not amenable to minutiae-based feature extraction methods. **a**-top Example of high-quality (tif101_8 from the database FVC20001 database) and **a**-bottom low-quality fingerprint (P5_Im4 from our in-house database), **b** binarization for both images, **c** thinning, and **d** the extracted minutiae . N.b. the extracted features from (top row), compared to those of the low-quality (bottom row) fingerprint image, which poses significant challenges. **a** Input, **b** binarized, **c** thinned, **d** features

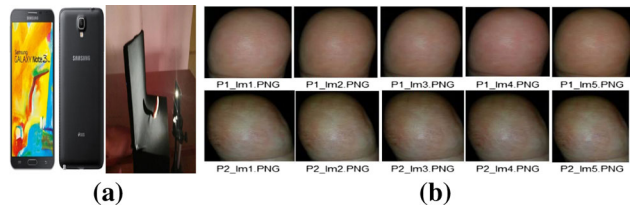

Fig. 2 The setup used in our smartphone fingerprint database (SPF) acquisition. **a** Samsung smartphone used in capturing images with lighting conditions and environment, and **b** examples of low-quality fingerprints obtained. Better viewed online and zoomed in. **a** Setup, **b** examples from the SPF database

the grayscale images without performing any steps such as binarization and/or thinning, (2) binary images, (3) machine learning techniques [19] or other techniques such as chain codes [24], run representation [25], (4) images after applying binary skeletons steps [12], which is one of the most widely used approaches, see [4] for a recent survey. Most of the previous studies with high-resolution fingerprint recognition used standard features called minutiae, which represents ridge terminations and ridge bifurcations. One of the main challenges facing fingerprint recognition systems is the absence of efficient ways to extract reliable features from low-quality fingerprint images. The minutiae-based methods rely mainly on minutiae detection to extract the features and can fail in low-quality scenarios. In Fig. 1(top row), we show the results of standard binarization and thinning steps and the extraction of minutiae on the high-quality image (tif101_8 from the database FVC20001[1]). Whereas when we applied the same steps using an image of low quality, we obtained inferior results as can be seen in the final minutiae features extracted in Figure 1(bottom row). Accordingly, we cannot detect the features using minutiae-based techniques for low-quality fingerprints, which necessitates using unusual methods to extract features reliably.

In this work, we introduce a new feature extraction method that does not depend on the detected minutiae and works well for recognizing low-quality fingerprints. The proposed method here depends on the energies provided by a fingerprint (here, we focus on the edges' energies of the fingerprint to approximate the energies possessed by a fingerprint) as these energies will be akin to magnetic energies, which attract iron filings, assuming that these iron filings are previously distributed uniformly over the fingerprint image. The coordinates of the attracted iron filings are saved as feature points to represent each fingerprint image. We apply machine learning techniques to test the efficiency of magnetic energy-based feature extraction in low-quality fingerprint images and perform various experiments. To the best of our knowledge, there is no low-quality fingerprint image database available for download; for this reason, a new in-house low-quality database is created to measure the performance of the proposed approach. The rest of this paper is dedicated to describing the image database created for the purpose of this study in Sect. 2, along with the details of the proposed method. Section 3 presents the experimental results, and conclusions are given in Sect. 4.

2 Description of the in-house database and feature extraction

2.1 Database details

Despite the availability of standard fingerprint databases, we could not find one with low-quality characteristics; therefore, we built our own in-house low-quality fingerprint database. We used a Samsung Galaxy Note 3 smartphone; see Fig. 2a, which is fast and easy to use. It provides best-resolution options and auto focus. In addition, it is equipped with Wi-Fi technology, which enables the users to easily access the captured images. Figure 2a clarifies the environment of our database collection. We called the in-house database smartphone fingerprint database (SPF), which has 680 different fingerprint images for 136 volunteers from different ages and genders; the ages of the subjects were in the range 14–60 years; majority of the subjects were males (101), and the remaining were females (35). The size of each image is 3096×41284, and the finger print is cropped manually and resized to 300×300 to speed up the process. Our smartphone fingerprint (SPF) database is available online[2].

We have designed a panel and a stand as tools to help the volunteer and the photographer to x their hands and to x the

[1] http://bias.csr.unibo.it/fvc2000/download.asp.

[2] https://www.mutah.edu.jo/biometrix.

Table 1 Environment and specifications of the used camera phone in our smartphone fingerprint (SPF) database collection

Measurement	Specified quality
Phone type and model	Samsung Galaxy Note 3
Sensor size	f/2.2 lens and a 1/3.2-inch sensor
Image file size	27 ± 3 MB
Image format	PNG
Flash	On
Background	Use black panel
Distance finger to phone	6–8 cm
Light conditions	Uniform outdoor environment
Image resolution	13 megapixels
Rotation degree	$[5 + 5]$

background in an easy way. The volunteers were asked to x their hands on this stand to constrain the rotation around $\pm$ 5 degrees vertically and to foster the clearance of fingerprint images as could as possible. Five images were taken for each volunteer's right thumb, and Table 1 summarizes the fingerprint-capturing environment parameters. The images were stored in portable network graphics (PNG) format. We used a file naming convention that encodes the number of volunteers followed by the image number, "VolunteerNumber_ImIndex." For example, the file (P16_Im2) represents the second image of thumb for the 16th volunteer. Figure 2b shows some examples of the annotated database.

2.2 Magnetic feature extraction with edges

In our proposed method, the strength of the fingerprint's edges is considered to approximate the energies of a fingerprint so as to obtain the most significant available features to identify a subject. Here, we opt for the Sobel filter, for its simplicity and common use. Perhaps finding the gradient is better, at least in terms of time consumed; however, tuning the proposed method is out of the scope of this paper.

The low-quality images were captured using a smartphone, then they were converted to grayscale images, and after that, we apply the Sobel filter to extract the energies in a fingerprint; see Fig. 3a, b, respectively. We next divide the image into a series of windows, with size (5×5) such that an iron filing is located on the center of each window/block; these appear with blue dots in Fig. 3c. Those iron filings are attracted to the points of the max magnetic energy within each block; i.e., each block has only one feature point, and chose a small size (5×5) of the block so as not to lose important features. Those iron fillings that did not move or moved to a point with a lower energy are removed, and those moved to high-energy points are stored forming a feature vector consisting of (x, y) points to describe the fingerprint. Here, we identify

Algorithm 1 Steps of our proposed method for feature extraction

Input: Fingerprint images
Output: Feature vectors with different lengths

1: Read a colored fingerprint image
2: Convert the colored image to grayscale image
3: Apply Sobel filter
4: Divide the image into 5×5 blocks
5: Put an iron filing on the center of each block
6: For each block, move the iron filing to the location of the max magnetic energy
7: For each iron filing,
8: **if** max (magnetic energy) < threshold)
9: Remove (iron filing)
10: **else** Store (x, y) location(iron filling)
11: **procedure** TRANSLATION
12: $x' = x - \min x,\ y' = y - \min y$
13: **procedure** SCALING
14: $X'' = \frac{x'}{\max x'},\ Y'' = \frac{y'}{\max y'}$
15: New vectors= matrix (X'', Y'')

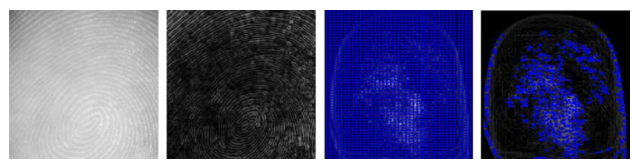

Fig. 3 Example of magnetic energy-based feature extraction method. **a** Grayscale version of an image from SPF database (P4_Im2), **b** after applying Sobel filter, **c** throwing iron filings uniformly, and **d** the attracted iron filings (features). **a** Input, **b** energies, **c** magnetic mask, **d** features

the low energy with a specific, predefined global threshold; see Fig. 3d. We conducted several experiments to find the best threshold for the database used, where our method achieved the best performance when using a threshold of 45 of energy, as shown in the next section.

It is worth mentioning that the length of the feature vector varies from one image to another, and therefore, needs special treatment when matching. To alleviate the effect of translation on the feature vector (x, y) points, we used, $x' = x - \min x$, $y' = y - \min y$, where x and y are the coordinates of the feature point, $\min x$ and $\min y$ are the minimum of x and y coordinates, respectively. Further, we used the min-max normalization $X'' = x'/\max x'$, $Y'' = y'/\max y'$ to minimize the effect of scaling. Algorithm 1 summarizes the steps of our proposed method for feature extraction.

2.3 Matching

In the previous steps, the features were converted into normalized 2D vectors, the x and y values of the points are concatenated to form 1D vectors, and then we use the K-nearest neighbors classier (KNN), with $k = 1$, and dynamic time warping (DTW) algorithm as similarity measure [21].

The DTW is capable of dealing with vectors regardless of their lengths. However, the major problem of this approach is the time complexity of DTW, which is $O(n^2)$, and this consumes a lot of time if n (the feature vector length) is a large value, which is the case when the energy threshold used is small (less than 50). In our experiments, the first two sets of experiments were applied using DTW and some experiments took significant computational time using leave-one-out cross-validation. The rest of experiments were conducted using some other machine learning method to control the problem of time complexity. Those methods require a fixed length of the feature vectors; therefore, we use the probability density function (PDF), which is based on the histogram of the (x, y) points (see experiments 3–7). This is done mainly to unify the varied lengths of all feature vectors, (see Sect. 3). We opt to use PDF instead of the frequencies of (x, y) points because it is an efficient technique for data scaling.

Finally, the x-PDF and the y-PDF of the (x, y) points are concatenated to generate one feature vector for each fingerprint. The histograms used by the PDFs use the following bin values: 5, 10, 15, …, 60. The PDF is calculated using

$$\mathrm{PDF}(I) = \frac{f(I)}{\sum f(I)}, \tag{1}$$

where I is the bin number, $\mathrm{PDF}(I)$ is the probability density function of I, and $f(I)$ is the frequency of I. Algorithm 2 summarizes the steps of matching and applying different classifiers. Figure 4 summarizes all steps of the proposed method.

Algorithm 2 Steps that used for matching

Input: The previous computed feature vectors
Output: Computed performance measures of the approaches

```
Read the vectors
Calculate the similarity using DTW
Apply KNN
Calculate performance measures for the first approach
procedure LOOP FOR BINS 5- - 60 WITH STEP SIZE 5
    Create PDFs
    Apply classifiers
    Calculate performance measures
```

3 Experimental results and discussions

In order to evaluate the proposed method, we conducted eight different sets of experiments, using two cross-validation schemes, 680-fold and fivefold cross-validation, and the results show three standard measures, namely accuracy, precision, and recall. In addition to the KNN-DTW, we utilize six different and widely used machine learning techniques [7]: Naïve Bayesian (NB), artificial neural networks (ANN), decision tree (DT), K-nearest neighbor (KNN), random forest (RF), and support vector machines (SVM). We implemented the machine learning techniques by using the well-known Weka tool [5]. In some experiments, we use principal component analysis (PCA) as a preprocessing step to reduce data dimensions.

3.1 Performance of classifiers and features

Leave-one-out cross-validation (680-fold) [2] is used with experiments 1 and 2, where the KNN-DTW classifier is used with $k = 1$, which involves the use of 1 observation as valida-

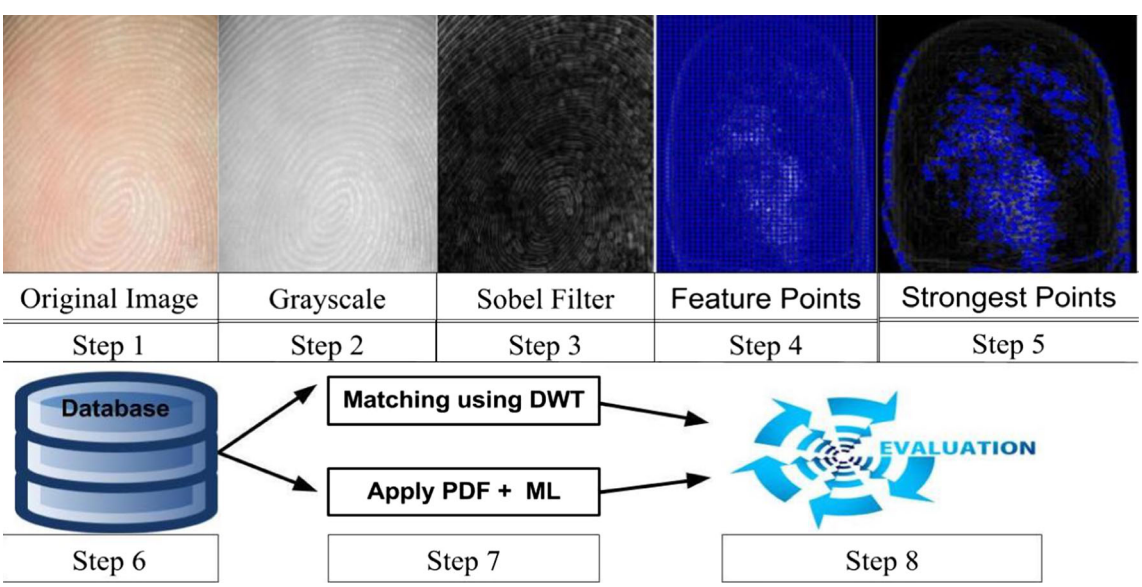

Fig. 4 Workflow of the proposed magnetic energy feature extraction method for the low-quality fingerprints

Table 2 Identification accuracy of KNN-DTW using different energy thresholds with leave-one-out cross-validation

Threshold	Accuracy	
	With boundary	Without boundary
30	88.63	91.50
35	90.20	92.65
40	93.17	93.38
45	**95.11**	**93.53**
50	94.30	91.62
55	94.12	90.00
60	92.35	88.82
65	90.59	84.71
70	89.41	82.62
80	83.24	70.88

We compare the accuracy with and without features on the boundary
Bold values indicate best result

tion subset and the remaining observations as a training data, while the rest of experiments used fivefold cross-validation with different classifiers.

3.1.1 Experiment set 1: KNN-DTW results using the whole features

In this experiment set, the normalized concatenated 1D feature vectors are used by the KNN-DTW without excluding any feature point such as those located on the boundary of the finger. We conducted several experiments to evaluate the identification performance using different energy thresholds between 30 and 80. The proposed method achieved satisfying results, as seen in Table 2, with the highest accuracy 95.11% achieved by using a threshold value 45. Thresholds from 30 to 55 result in higher accuracies, but allow for consuming more time as the feature size increases, while the higher thresholds allow for consuming considerably less time as the feature vector size decreases, but deteriorate the accuracy, with the lowest accuracy when using the highest threshold 80. This is due to the removed points of lower energy, which might hold distinctive information.

3.1.2 Experiment set 2: KNN-DTW results with excluding the boundary features

This experiment set is similar to Experiment set 1, with one difference which is that we exclude the features that are located on the boundary of the fingerprint, attempting to reduce the size of the feature vector. The size of the feature vector at energy threshold = 30 is in the range of 3000–4500, when we ignored the feature points that located on the boundary, the sizes of the feature vectors are reduced to be around 1000 on average, and this makes the system about 3 times faster. Knowing that these points keep only the shape information of the finger, which is almost the same for all. By comparing the results in Table 2, we see that the accuracies slightly deteriorated when ignoring the boundary's information, which indicates the existence of some important features on the boundaries; however, the algorithm becomes considerably faster due to the reduced size of the feature vector. Figure 5 shows a comparison between experiments 1 and 2, which shows the effect of keeping the boundary information on the overall accuracy using different energy thresholds. The next Experiment sets (3–8) were conducted on a fixed-sized feature vector determined by the number of bins used, and we will show the performance of the proposed method using different number of bins 5, 10, 15,…60.

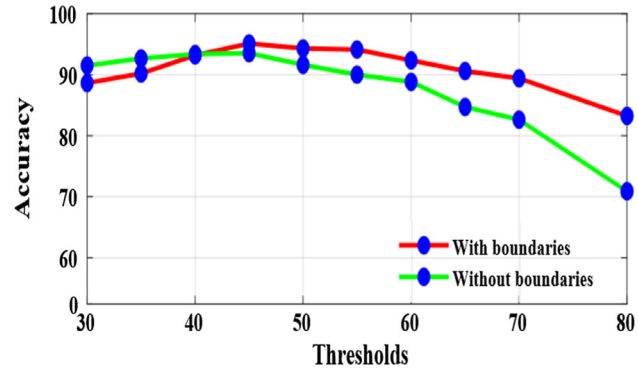

Fig. 5 KNN-DTW results with/without boundaries

3.1.3 Experiment set 3: KNN classier with PCA (fivefold)

Here, we use different bins to create two PDFs for each, one for the x values and the other for the y values, which both concatenated to form the final feature vector for each number of bins. After that, we apply the PCA to reduce data dimensionality, and these data were classified using the Weka's KNN with Euclidean distance as similarity measure; here, we use fivefold cross-validation because each subject in the database has only five fingerprint images. From Table 3, we note that the best accuracy is obtained when using (10–15) bins; it is also interesting to note that the accuracy decreases as the number of bins increase; this is due to the small resulting probability from equation (1), where the probability for each bin number decreases when the number of bins increases, forming a feature vector of very small numbers, which makes the classifier performs poorly.

3.1.4 Experiment set 4: KNN classier without PCA (fivefolds)

This experiment set is similar to Experiment set 3; one exception is ignoring the PCA step. As can be noted from Table 3, the best results were obtained when using (10–15) bins. It can be noted that the PCA is not a good choice for our PDF

Table 3 KNN classier with/without PCA (fivefold)

Bins	Accuracy	Precision	Recall
5	78.38/79.85	79.4/79.9	78.4/79.9
10	**83.08**/85.59	**84**/85.9	**83.1**/85.6
15	79.70/**85.89**	80/**87.50**	79.7/**85.90**
20	72.64/ 85.14	73.4/86.4	72.6/ 85.1
25	64.55/84.41	65.0/ 85.00	64.6/84.44
30	55.88/81.61	56/81.70	55.9/81.16
35	46.47/80.73	47/81	46.5/80.7
40	42.94/78.8	43/78.9	42.9/78.8
45	34.85/76.91	35/82.4	34.9/76.9
50	28.67/73.67	29/74.00	28.7/73.7
60	21.17/69.11	22.2/70	21.2/69.1

Bold values indicate best result

Table 4 KNN classier with/without PCA (680-fold)

Bins	Accuracy	Precision	Recall
5	78.38/78.82	79.5/79.1	78.4/78.8
10	**81.47/85.88**	**82**/86.8	**81.5/85.9**
15	79.62/85.73	80/**87.4**	79.3/85.7
20	72.05/84.55	72 /86	72.1/84.6
25	62.79/84.11	63/85.9	62.8/84.1
30	53.67/81.32	54/83.2	53.7/81.3
35	45.29/79.85	46/80	45.3/79.9
40	40.44/78.82	41/79	40.4/78.8
45	33.32/76.32	33 /77.0	33.2/76.3
50	27.20/72.5	27 /73	27.2/72.5
60	20/68.8	20/69	20/68.8

Bold values indicate best result

data, particularly when the number of bins is large, as the features become very small.

3.1.5 Experiment set 5: KNN classier with PCA (680-fold)

Similar to Experiment set 3, but here we evaluate the proposed method using leave-one-out cross-validation. i.e. 680-fold, where we apply KNN on the data that is preprocessed by PCA. As can be noted from Table 4 and comparing to Table 3, the results are almost the same with a slight decrease in accuracy, which shows the stability of the features extracted by the proposed method.

3.1.6 Experiment set 6: KNN classier without PCA (680-fold)

This is similar to Experiment set 3, but here we evaluate the proposed method using leave-one-out cross-validation. That is, 680-fold, we apply KNN without data preprocessing. As can be seen from Table 4, using ten bins (again) allows for higher accuracy. Also, we see that the number of folds has no clear impact on the final accuracy, which (again) indicates the stability of the features extracted by the proposed method.

Table 5 Comparison of classifiers using $10x$, $10y$ bins without PCA (fivefold)

Classier	Accuracy	Precision	Recall
Support vector machine	65.59	66	65.6
Artificial neural network	82.65	83	82.6
K-nearest neighbor	**85.59**	**85.9**	**85.6**
Naïve Bayes	64.41	84.0	64.4
Decision tree	27.2	27.2	27.2
Random forest	82.64	83	82.6

All classifiers used their own default parameters
Bold values indicate best result

Table 6 Comparison of classifiers after features selection without PCA (fivefold)

Classier	Accuracy	Precision	Recall
Support vector machine	53.97	54.3	54
Artificial neural network	75.74	76	75.7
K-nearest neighbor	**79.26**	**80**	**79.3**
Naïve Bayes	61.32	61.4	61.3
Decision tree	29.26	29.2	29.3
Random forest	76.02	76.2	76

Bold values indicate best result

3.1.7 Experiment set 7: comparison of classifiers using 10*x*, 10*y* bins without PCA (fivefold)

Having seen that the ten-bin choice is the best in terms of accuracy (ten bins means 20 features, as we have ten probabilities for the xs and the same for the ys), we use this choice to further evaluate our method using six different classifiers. With fivefold cross-validation, the results are recorded in Table 5, and we note that the best results were obtained when implementing KNN with accuracy 0.8559 followed by ANN with accuracy 0.8265, and the worst accuracy was obtained with the use of decision tree with 0.272. It is expected to see these results decreasing dramatically comparing to those of the KNN-DTW because of losing information by categorizing the data to a finite number of bins.

3.1.8 Experiment set 8: comparison of classifiers after feature selection without PCA (fivefold)

Features selection is a process that orders features according to their distinctive strength, and then it uses a specific searching strategy to select the 'strongest' features and remove the 'weakest' ones. Considering the $10x$, $10y$ bins = 20 features, we use the Weka's 'CfsSubsetEval evaluation method with 'BestFirst as a searching strategy to select the 'strongest' 12

Table 7 Comparison between previous related studies using various medium- to low-quality fingerprint imageries and our proposed approach

Refs.	Capturing tool	Performance	Subjects
[22]	Nokia N8	Accuracy = 97.4%	25
	iPhone 4	Accuracy = 98.43%	
	Samsung S1	Accuracy = 97.25%	
[18]	Webcams from Apple, Microsoft, Philips	FAR = 0.18% FRR = 10.29%	6
[16]	Samsung Galaxy I, Nokia N8	Highest EER 23.6%	25
[8]	Canon PowerShot Pro1 (8 mp)	Accuracy = 95.44%	–
[29]	Cannon PowerShot A75, resolution 110dpi	PCA 3.16% LDA 1.74%	43
[14]	Mobile phone camera	Controlled Environment	13
		Detection rate = 82.98%	
		Uncontrolled Environment	
		Detection rate = 78.21%	
[15]	Nokia 6700 classic (5 mp), HTC magic (3.5 mp)	Accuracy = 77.95%	13
[10]	Webcam	FAR = 0.042% FRR = 3.92%	1200 images
[23]	Apple iPhone5 (8 mp)	EER 3.10%	5100 images
[26]	Samsung Nexus S, Galaxy Nexus	EER < 20%	41
[17]	Nokia N8, iPhone 4, Samsung Galaxy S	Highest EER 35.3%	2100 fingers
Our	Samsung Note III	Accuracy = 94.2%	136

FAR false acceptance rate, *FRR* false rejection rate, *EER* equal error rate, *PCA* principle component analysis, *LDA* latent Dirichlet allocation, *mp* mega pixels

features. Then, we apply the same six classifiers. The results shown in Table 6 indicate that the feature selection method used affects the results negatively, and this is expected because the feature vector has already lost some information by binning, so any further feature extraction/selection would affect the results by decreasing the accuracy of the classifier used.

3.2 Comparison with other low-quality fingerprints classifications

Finally, we compare our results with the results of some previous studies that were conducted for fingerprint recognition using digital images acquired by commercial cameras, smartphones, and webcams. The comparison is shown in Table 7. These studies do not use high-quality scanners for image acquisition; for example, [8] and [29] used a Canon digital camera, whereas [18] and [10] used webcams, and the rest are based on imagery taken from smartphone digital cameras. It is interesting to note that the results of [22] outperform other smartphones results with accuracy of 98.43%; this is expected as the number of subjects forming the databases is very small (25 subjects). Overall, we see that the proposed method achieves satisfying results with the limited quality of the digital images used comparing to the reported results from the recent literature. However, this comparison is not perfect since each study used different database, hardware, number of subjects, and number of images per subjects. Further, automatic image quality assessment techniques are important in comparing such mobile applications (low cost), and security [9,27].

4 Conclusions

We propose a new approach for recognizing individuals using low-quality fingerprint images, along with a new low-quality database called smartphone fingerprint database (SPF). Although the images we used were of low quality, our experiments have shown a great potential. We used edges-based magnetic features that work well in this scenario. Dynamic time warping (DTW) was used to compare each features vector with the others, and we evaluated the performance of our system based on KNN classier. We used the histogram of probability density functions (PDFs) to prepare the data for different classifiers. Our results show that the edges contain important information for improving the quality of fingerprint system, particularly, with low-quality images. We proved empirically that using the whole features from all components of the image resulted in better accuracy results. We also noticed that the use of original features achieved better results than that of preprocessed data; we noticed that the results were affected negatively by using PCA as a preprocessing technique. In the future, a hybrid method for feature extraction can be used by combining minutiae and edges. In addition to this, we can use sophisticated enhancement and edge detection techniques to improve the magnetic

features capability [1,30]. Moreover, we will enlarge our SPF dataset up to 1000 volunteers with ten samples for each one to get more stable results.

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